

Nathan Kwan-Ho Ng

Ph.D. Candidate, Computer Science
University of Massachusetts Amherst

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EDUCATION

Ph.D. in Computer Science

09/2019 – 09/2026 (Expected)

University of Massachusetts Amherst

Advisor: Prof. Prashant Shenoy

B.Sc. in Computer Science

09/2014 – 06/2018

Hong Kong Baptist University

Advisor: Prof. Byron Choi

RESEARCH INTERESTS

My research lies broadly in distributed systems and systems for machine learning, with a current emphasis on AI systems across the device–edge–cloud continuum. I design, build, and optimize systems and resource-management techniques that deliver predictable low-latency performance, covering both average-case and tail-latency behavior, for emerging AI applications.

PUBLICATIONS & PREPRINTS

Tuning the Tail Latency of Distributed Queries Using Replication

Nathan Ng, Hung Le, Prashant Shenoy, Marco Serafini

Under review at the *Distributed and Parallel Databases Journal (DAPD)*. arXiv:2212.10387

Collaborative Processing for Multi-Tenant Inference on Memory-Constrained Edge TPUs

Nathan Ng, Walid Hanafy, Prashanthi Kadambi, Balachandra Sunil, Ayush Gupta, David Irwin, Yogesh Simmhan, Prashant Shenoy

International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT'26)

Best Paper Candidate

To Offload or Not To Offload: Model-driven Comparison of Edge-native and On-device Processing in the Era of Accelerators

Nathan Ng, David Irwin, Ananthram Swami, Don Towsley, Prashant Shenoy

ACM/SPEC International Conference on Performance Engineering (ICPE'26)

TailClipper: Reducing Tail Response Time of Distributed Services Through System-Wide Scheduling

Nathan Ng, Abel Souza, David Irwin, Don Towsley, Prashant Shenoy

ACM Symposium on Cloud Computing (SoCC'24)

Collaborative Inference in Resource-Constrained Edge Networks: Challenges and Opportunities

Nathan Ng, Abel Souza, Suhas Diggavi, Tarek Abdelzaher, Don Towsley, Prashant Shenoy

IEEE Military Communications Conference (MILCOM'24), IoT for Adv. Env. Workshop

Unlocking Efficiency: Understanding End-to-End Performance in Distributed Analytics Pipelines

Abel Souza, Nathan Ng, Tarek Abdelzaher, Don Towsley, Prashant Shenoy

IEEE Military Communications Conference (MILCOM'23), IoT for Adv. Env. Workshop

FGreat: Focused Graph Query Autocompletion

Nathan Ng, Peipei Yi, Zhiwei Zhang, Byron Choi, Sourav S. Bhowmick, Jianliang Xu
IEEE International Conference on Data Engineering (ICDE'19), Demonstration Track

Answering the Why-Not Questions of Graph Query Autocompletion

Guozhong Li, **Nathan Ng**, Peipei Yi, Zhiwei Zhang, Byron Choi
International Conference on Database Systems for Advanced Applications (DASFAA'18)

PATENTS

Dynamic user profile management

Nathan Ng, Tung Mai, Thomas Greger, Kelly Quinn Nicholes, Antonio Cuevas, Saayan Mitra, Somdeb Sarkhel, Anup Bandigadi Rao, Ryan A. Rossi, Viswanathan Swaminathan, Shivakumar Vaithyanathan
US Patent US12,339,916B2. Published June 2025.

Dynamic user profile projection

Nathan Ng, Tung Mai, Thomas Greger, Kelly Quinn Nicholes, Antonio Cuevas, Saayan Mitra, Somdeb Sarkhel, Anup Bandigadi Rao, Ryan A. Rossi, Viswanathan Swaminathan, Shivakumar Vaithyanathan
US Patent US12,130,876B2. Published Oct 2024.

AWARDS AND HONORS

Best Paper Candidate <i>DCOSS-IoT'26</i>	2026
Donald F. Towsley Graduate Scholarship in Computer Science <i>University of Massachusetts Amherst</i>	2026
James Kurose Scholarship in Computer Science <i>University of Massachusetts Amherst</i>	2020
Undergraduate Overseas Summer Research Scholarship <i>Hong Kong Baptist University</i>	2018
Outstanding Student Scholarship <i>Hong Kong Baptist University</i>	2017
President's Honor Roll <i>Hong Kong Baptist University</i>	2016–2018

RESEARCH EXPERIENCE

Research Assistant , <i>University of Massachusetts Amherst</i> Advisor: Prof. Prashant Shenoy	09/2022 – Present
- Developed SwapLess, a system enabling multi-tenant collaborative inference on memory-constrained Edge TPUs. - Developed an analytic-model-driven resource manager for adaptive inference offloading in edge environments. - Developed TailClipper, a queueing-theory-inspired scheduler that reduces tail latency of distributed applications.	
Incoming Research Scientist Intern , <i>Adobe</i> Mentor: Dr. Saayan Mitra	06/2026 – 09/2026 (Expected)
Developing privacy-preserving LLM inference offloading strategies within Adobe Acrobat.	
Research Scientist Intern , <i>Adobe</i> Mentor: Dr. Tung Mai	05/2022 – 08/2022

- Developed an Observability Data Management System deployed in production for Adobe engineering teams.
- Developed a URL knowledge extraction system using Random Forest to improve metrics reporting.

Data Science & Machine Learning Intern, Adobe

05/2021 – 12/2021

Mentor: Dr. Tung Mai

- Proposed a cache admission algorithm using a predictive statistical model to forecast customer revisits.
- Proposed dynamic Cloud-Edge user profile replication algorithms to maximize cache hit rates.

Student Research Collaborator, Meta

01/2021 – 05/2021

Mentor: Dr. Ritwik Yadav

- Benchmarked performance of RocksDB and LiveGraph using LinkBench on AWS EC2 instances.
- Collaborated on a potential deployment of LiveGraph as a Database-as-a-Service model at Meta.

Research Assistant, University of Massachusetts Amherst

09/2019 – 05/2022

Advisor: Prof. Marco Serafini

- Proposed data replication algorithms to satisfy user-defined tail latency bounds for distributed queries.
- Developed a distributed graph query processing middleware that coordinated query execution across nodes.

Research Assistant, Hong Kong Baptist University

09/2018 – 08/2019

Advisor: Prof. Byron Choi

- Proposed algorithms for efficient indexed maximum common subgraph mining in graph databases.

Undergraduate Student Researcher, Hong Kong Baptist University

09/2017 – 08/2018

Advisor: Prof. Byron Choi

- Proposed ranking and search algorithms to address the ‘why-not’ questions in graph query autocompletion.

Undergraduate Research Intern, Nanyang Technological University

05/2017 – 08/2017

Advisor: Prof. Sourav S. Bhowmick

- Developed FGreat, an interactive graph query autocompletion system that assists query formulation.

TEACHING EXPERIENCE

CS 377: Operating Systems, Undergraduate Teaching Assistant

Spring 2025

University of Massachusetts Amherst

- Led lab sessions for 50 undergraduates covering OS concepts and hands-on assignments.

CS 677: Distributed & Operating Systems, Graduate Teaching Assistant

Spring 2023, 2024

University of Massachusetts Amherst

- Held office hours, graded coursework, and provided Piazza support for students.

CS 532: Systems for Data Science, Graduate Teaching Assistant

Fall 2019, 2022, 2023

University of Massachusetts Amherst

- Designed and led lab sessions on Hadoop and Spark for 120 graduate students.

CS 2007: Object-Oriented Programming, Undergraduate Teaching Assistant

Spring 2018

Hong Kong Baptist University

- Assisted students with hands-on debugging and Java programming exercises.

CS 1005: Essence of Computing, Undergraduate Teaching Assistant

Fall 2017

Hong Kong Baptist University

- Assisted freshman students with programming exercises and debugging.

MENTORING

Sribatscha Maharana, B.Sc. and M.S. in Computer Science (UMass'27)	Spring 2026
Naga Surya Prathik Vunnava, M.S. in Computer Science (UMass'26)	Spring 2026
Balachandra Devarangadi Sunil, M.S. in Computer Science (UMass'26)	Fall 2025
Ayush Gupta, M.S. in Computer Science (UMass'26)	Fall 2025
Khai Nguyen, M.S. in Computer Science (UMass'25)	Spring 2025
Tanay Joshi, M.S. in Computer Science (UMass'25)	Spring 2025
Rishi Patel, B.Sc. in Computer Science (UMass'25)	Spring 2025

SERVICE & OUTREACH

External Reviewer	Fall 2025
<i>ACM/IEEE Symposium on Edge Computing (SEC'25)</i>	
Shadow PC Member	Fall 2025
<i>The European Conference on Computer Systems (EuroSys'26)</i>	
Instructor	Summer 2024, 2025
<i>UMass Turing Program</i>	

REFERENCES

Prashant Shenoy Distinguished Professor <i>University of Massachusetts Amherst</i> shenoy@cs.umass.edu	Don Towsley Distinguished Professor <i>University of Massachusetts Amherst</i> towsley@cs.umass.edu	David Irwin Professor <i>University of Massachusetts Amherst</i> irwin@ecs.umass.edu
Hui Guan Assistant Professor <i>University of Massachusetts Amherst</i> huiguan@cs.umass.edu	Tung Mai Senior Research Scientist <i>Adobe</i> tumai@adobe.com	